

# Project - Low Level Design

**on**

**Autonomous Energy**

**Researcher Agent**

Course Name: Agentic AI

***Institution Name:*** Medicaps University – Datagami Skill Based Course

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*Academic Year: 2026*

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# 1. Introduction

The **Autonomous Energy Researcher Agent** is an AI-powered multi-agent system designed to automate the collection, analysis, validation, and storage of intelligence related to the global energy sector.

It uses **LangChain**, **CrewAI**, and **Large Language Models (LLMs)** to independently search trusted sources, extract relevant data, synthesize insights, and store findings in a reusable knowledge base.

The platform helps reduce manual research time and provides fast, accurate, and structured market intelligence on:

- Energy market trends
  - Policy and regulatory changes
  - Renewable energy developments
  - Oil & Gas intelligence
  - Pricing updates
  - Technology breakthroughs
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## 1.1 Scope of Document

This Low-Level Design document covers:

- Internal architecture of the multi-agent system
  - Agent responsibilities and coordination logic
  - Search and data extraction workflow
  - Information validation mechanisms
  - Knowledge repository structure
  - APIs and tools used
  - Design considerations
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## 1.2 Intended Audience

This document is intended for:

- Developers building agentic AI systems
  - Energy analysts using AI for market research
  - Architects designing autonomous AI pipelines
  - Organizations implementing research automation
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## 1.3 System Overview

The system follows a **Manager–Worker Multi-Agent Architecture**.

### Core Agents:

#### 1. Research Manager

Breaks the user query into smaller research tasks.

#### 2. Researcher Agents

Search web sources, collect reports, scrape websites, and extract facts.

#### 3. Validation Agent

Cross-verifies claims using multiple trusted sources.

#### 4. Knowledge Architect

Formats final results and stores them in a searchable repository.

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## Technology Stack

- Python
- LangChain (ReAct Agents)

- CrewAI
- Flask API
- OpenAI / Anthropic APIs
- Serper Search API
- Markdown/Text Storage

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## 2. Application Design

The application uses a **Modular Agentic Framework**.

Instead of relying only on static LLM knowledge, it actively performs:

- Real-time web research
- Multi-step reasoning
- Source validation
- Structured summarization
- Persistent memory storage

This allows accurate responses to current market conditions.

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### 2.1 Workflow Diagram

User Query



Research Manager



Task Distribution



Researcher Agents



Web Search + Scraping



Entity Extraction



Cross Validation



Summary Generation



Knowledge Repository

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## 2.2 Process Flow

### Step 1: Query Input

Example:

**"Global Offshore Wind Capacity Forecast for 2026"**

### Step 2: Task Breakdown

Manager converts into tasks:

- Search current offshore wind installations
- Find government capacity targets
- Extract region-wise capacity data

### Step 3: Search Execution

Researchers use tools to search:

- Government websites
- Industry reports
- News portals
- Energy databases

### Step 4: Extraction

System extracts:

- Capacities (MW / GW)
- Dates
- Geography

- Pricing
- Policy changes

## Step 5: Validation

Checks duplicate or conflicting claims.

## Step 6: Final Summary

Creates professional report.

## Step 7: Persistence

Stores output in repository.

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## 2.3 Information Flow

User Request



Intent Understanding



Task Planning



Web Retrieval



Filtering & Deduplication



Fact Extraction



Insight Generation



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## 2.4 Component Design

### 1. Config Module

Stores:

- API Keys
  - Search credentials
  - Model settings
  - Storage path
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### 2. Search Intelligence Engine

Ranks sources based on authority:

High Priority:

- Government websites
- IEA / EIA Reports
- Bloomberg / Reuters
- Official company filings

Low Priority:

- Blogs
  - Forums
  - Unverified pages
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### 3. Energy Entity Extractor

Converts raw text into structured JSON:



## 4. Synthesis Engine

Creates final analyst-grade summary:

- Removes duplicates
  - Combines multiple sources
  - Highlights conflicting claims
  - Adds source citations
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## 5. CrewAI Orchestration Layer

Handles:

- Agent creation
  - Role definitions
  - Tool assignment
  - Sequential workflows
  - Task dependencies
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## 6. Knowledge Repository

Stores reports as:

2026\_Offshore\_Wind\_Report.md

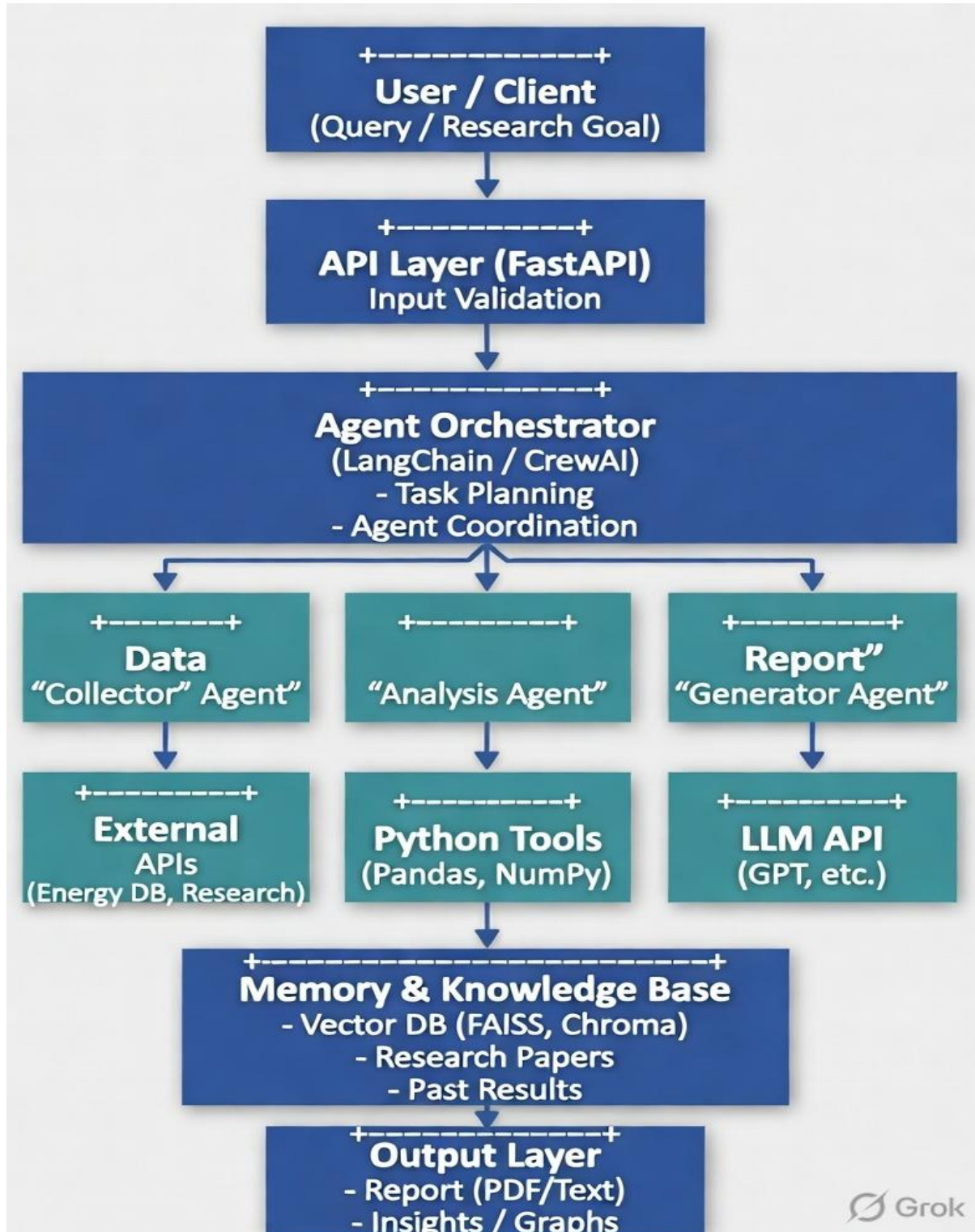
Hydrogen\_Market\_Analysis.txt

Gas\_Pricing\_Q1\_2026.md

Supports future RAG systems.

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## 2.5 Architecture Diagram



## 2.6 Key Design Considerations

### **Search Depth**

Top 5 vs Top 10 results.

### **Accuracy**

Every claim linked with source URLs.

### **Cost Optimization**

Dynamic LLM usage based on query size.

### **Storage Strategy**

Append vs overwrite logic.

### **Duplicate Detection**

Prevents repeated reports.

### **Security**

Protects API keys and local files.

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## 2.7 API Catalogue

Tool / API

Purpose

SerperDevTool      Real-time Google search

ScrapeWebsiteTool      Reads webpage content

FileReadWriteTool      Saves reports

OpenAI API      Reasoning + extraction

CrewAI      Multi-agent execution

Flask API      External access

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### 3. Future Enhancements

- PDF / PowerPoint report generation
  - Dashboard with charts
  - SQL vector database
  - Live market alerts
  - Multi-language research
  - Voice-based querying
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## 4. Conclusion

The Autonomous Energy Researcher Agent transforms traditional manual research into an intelligent autonomous pipeline.

It combines:

- Real-time web intelligence
- Multi-agent reasoning
- Data validation
- Automated reporting
- Long-term knowledge retention

This makes it highly useful for analysts, consulting firms, and energy enterprises.